wheat sourdough

White wheat flour + 25% whole wheat flour. Stiff starter, lievito madre at 43% hydration — my standard, super-active one, the starter I baked bread and pastry on at my bakery. Using a different starter? Don't worry, the recipe can be adapted! Further down the recipe there's a calculator that converts it to a liquid starter and to any amount of flour and number of loaves. Dough hydration in this recipe is 75%. Sounds scary, it's simple.

75%	43%	1 night
HYDRATION	STARTER	COLD PROOF



white + whole wheat

- [ ] scale — MUST
- [ ] bowl + spoon — MUST
- [ ] basket *or bowl + towel* — SWAP
- [ ] cast iron *or roasting bag* — SWAP
- [ ] blade *or sharp knife* — SWAP
- [ ] parchment + mitts — MUST
- [ ] rack *or the oven rack* — SWAP
- [ ] thermometer — NICE

what you need

Tick what you already have before you start. Almost everything has an easy swap — except the scale.

what goes in

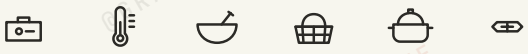
ingredient	small loaf 600 G	big loaf 1150 G
starter · 43%	71 g	137 g
whole wheat flour	85 g	163 g
white bread flour	204 g	391 g
water	233 g	446 g
salt	7 g	13 g

Starter 43% = 21% of the flour. Inside it: 50 g flour + 21 g water (600 g loaf), 96 g flour + 41 g water (1150 g loaf).

11%+
PROTEIN · WHITE FLOUR

10%+
PROTEIN · WHOLE WHEAT

Mine is Caputo Nuvola, 12.5%. Tools that matter: a scale, cast iron or any covered pot, and a blade for scoring.



liquid starter 100%?

a	Swap so the flour and water in the dough stay the same.
b	600 g: starter 100 g (not 71), water 204 g (not 233). Everything else the same.
c	1150 g: starter 192 g (not 137), water 391 g (not 446). Everything else the same.
d	Any other starter: it must carry 50 g / 96 g of flour; cut the dough water by whatever it brings above 21 g / 41 g. Or let the calculator do it.

01 mix & walk away

a	Weigh the water. Tear the starter into pieces — it gets to work faster.
b	White flour in, then whole wheat.
c	Mix with a spatula, just to wet the flour. No gluten development yet.
d	Cover and leave for autolyse.

30–50
MIN AUTOLYSE

just wet the flour →

▶ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s1

02 salt & the important part

a Salt in. Now knead properly.

b Taste from two spots. One bland, one salty? Keep kneading.

c At the end, measure the dough temperature. Mine was 27 °C.

d In this recipe my working range is 23–25 °C. Above 26 °C fermentation in my process runs ahead of the gluten, so I cool the dough — 30 min in the fridge.

2%	26°	23–25°
SALT OF FLOUR	ABOVE IT — COOL DOWN	MY WORKING RANGE

dough too warm? use cold water next time →

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s2

03 back in the optimal zone

a Out of the fridge: the dough is 25 °C. A great temperature for fermentation in the apartment.

b Next fold: wet hands, stretch one side up and fold it over, turn the bowl, repeat on all four sides.

c Cover and leave for 30 min until the next fold.

25°	23–25°
AFTER THE FRIDGE	MY WORKING RANGE

a calm dough, right on track →

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s3

04 folds & waiting

a A fold every 30 minutes, four in total.

b By the third one you see bubbles. A good sign.

c On the fourth, check the gluten: stretch a piece thin. A window like this is what we want.

d Now wait until the dough grows by 50–60%. Mine took about an hour and a half. And please forgive the sunset light in the last shots! I beeeg you

4
FOLDS · EVERY 30 MIN

+50–60%
VOLUME, NOT TIME

watch the dough, not the clock →

▶ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s4

05 shape & say good night

a	Preshape: round it, tightening the top. Rest 20–30 min.
b	Final shape: I do folds and swaddling. Any method you like works.
c	Another 20–30 min proof, then into the fridge overnight, covered.
d	My fridge is about 7 °C. At that temperature fermentation keeps going, so the overnight timing is for my fridge and this process: in a colder fridge the dough can stay longer, in a warmer one bake it earlier.

20–30
MIN REST · ×2

~7°
MY FRIDGE, OVERNIGHT

tension on top, not force →

▶ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s5

06 good morning. let's bake

a	Morning. The dough looks like this — time to bake.
b	Turn it out onto a mat, score on the side.
c	The dutch oven has been heating for an hour at 250 °C.
d	Lid on — 20 min. Lid off, down to 220 °C — another 20–25.

250°
20 MIN, LID ON

220°
20–25 MIN, LID OFF

the score opened a bit weakly? no worries! with a big loaf that's completely normal! →

▸ watch this step: aquadecastelo-cloud.github.io/grain-of-taste/recipes/wheat-sourdough.html#s6

TIMING AT A GLANCE

WATCH THE DOUGH, NOT THE CLOCK

a	Autolyse 30–50 min → salt, knead → temperature check: above 26 °C, cool to 23–25 °C
b	4 folds, every 30 min → bulk until +50–60%
c	Preshape, rest 20–30 min → shape, proof 20–30 min → fridge overnight (~7 °C)
d	Bake 250 °C lid on 20 min + 220 °C lid off 20–25 min

Timing is approximate. A liquid starter may run a little faster or slower than lievito madre.

TROUBLESHOOTING

IF SOMETHING WENT WRONG

problem	possible cause
The loaf spread flat	*Not enough structure developed, or overproofed*

Dense crumb	*Underfermented*
Too sour	*Starter ripeness and how the fermentation went*
Crust too thick	*Baking mode and moisture loss*
The loaf barely opened	*Shaping, proofing or scoring*

NUTRITION

APPROXIMATE, PER 100 G OF BREAD

228 KCAL	8.4 g PROTEIN	1.2 g FAT	47 g CARBS

a A slice of about 40 g is around 90 kcal. The whole loaf is about 2230 kcal · protein 82 g · fat 11 g · carbs 458 g.

b Only the flour counts: 487 g white (391 g in the dough + 96 g inside the starter) and 163 g whole wheat. Water and salt have no calories.

c Estimate for a loaf that lost about 15% of its weight in the oven (1150 g of dough → about 980 g of bread). Weigh yours and divide the whole-loaf numbers by its weight for an exact figure.

d Carbs include about 2.5 g of fibre. Salt: about 1.3 g per 100 g.

THE RESULT

CRUST · CRUMB · SIDE



the ear



listen to that crust



one hour later

and... here it is

Here is our bread. The score opened a bit weakly — that's the volume. In homemade bread that's not the point. The point is taste, craft and love.



taste, craft & love